

```
1. #include <bits/stdc++.h>
2. //DOCTOR_PATIENT
3. using namespace std;
4.
5. bool solution(vector<int> &A, vector<int> &B, int S) {
6.     set<pair<int, int>> slots; //set to store pairs of co-ordinates, because sets don't contain
    duplicate values
7.     int x = A.size(); //to be used in place of vector size thingy
8.     if(x > S) return false; //to check whether the arrays are larger than the doctor's available time
9.     //The magic begins...
10.    for(int i=0; i< x;i++){
11.        pair<int,int> current = make_pair(A[i], B[i]); //create a co-ordinate pair for each index
12.        pair<int,int> currRev = make_pair(B[i], A[i]); //create a reverse co-ord pair if the
    "current" pair won't work
13.        if (slots.find(current) == slots.end()) slots.insert(current); //insert non-reversed pair if
    it doesn't already exist
14.        else if (slots.find(currRev) == slots.end()) slots.insert(currRev); //insert reversed pair if
    the non-reversed pair already exists
15.        else return false; // if both pair already exists, then the slot is already full for the
    given time and so it should already be returned as false, as there won't be a unique available time
16.    }
17.    return true; //return true if everything works out great!
18. }
19.
20. int main()
21. {
22.     vector<int> a={1,1,1};
23.     vector<int> b={2,2,1};
24.     int slot=3;
25.     cout<< solution(a,b,slot);
26.     return 0;
27. }
28.
```